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Test of Hypothesis and Significance

Lecture - V

Statistical Decisions:

Very often in practice we are called upon to make decision about populations on the basis of sample information. Such decisions are called statistical decision.

Ex: 1 Suppose in a class there are 100 students and this constitutes the population set. Let the average percentage of marks is 70%. Now we ~~cons~~ select any 20 number of students out of them. This set is called the sample set. The average percentage of marks for these 20 student is 71% say.

Therefore the population average $\approx 70\%$
and the sample average $\approx 71\%$.

The difference between them is 1%.
We are to determine how significant the difference is.

Ex: 2 We may wish to decide on the basis of sample data whether a new serum is really effective in curing a disease.

Statistical Hypothesis: In attempting to reach decisions, it is useful to make assumptions or guess about the populations involved. Such assumptions which may or may not be true is called statistical Hypothesis.

• Let us consider an example: Suppose we want to know whether consumption of tobacco causes cancer. Or where whether there is any correlation between consumption of tobacco and cancer.

To do that we study two groups of human, who has consumed tobacco and who has not consumed tobacco.

~~If we find that there~~

The hypothesis - 1: There is no evidence to prove that the cause of cancer is related with the consumption of tobacco. i.e. the difference of statistical investigations between the two groups are zero or null. This is called the Null hypothesis.

hypothesis 2: There is significant evidence that to prove that the cause of cancer is related to the consumption of tobacco. This is called the alternative hypothesis.

Ex: Given a coin we construct two hypothesis -

H_0 : The coin is fair

H_1 : The coin is not fair.

To test the hypothesis, the following decision rule is adopted.

(i) Accept the null hypothesis H_0 if in a single sample of 100 tosses the number of head (tail) is between ~~60 and 80~~ 40 and 60 inclusive.

(ii) Reject the null-hypothesis or accept the alternative hypothesis otherwise.

Test of hypothesis and significance:

If on the supposition that a particular hypothesis

is true we find that results observed in a random sample differ markedly from those expected under the hypothesis on the basis of pure chance, we would say that the ~~by~~ observed differences are significant and we would be inclined to reject the hypothesis.

Ex: If 20 tosses of a coin yields 16 heads, we would be inclined to reject the hypothesis that the coin is fair. Although it is conceivable that we might be wrong.

Procedures that enable us to decide whether to accept or reject hypothesis or to determine whether observed sample differ significantly from expected results are called tests of hypothesis, tests of significance or decision rule.

Typ I and Typ II errors

if we reject a hypothesis when it happens to be true, we say that the error is type I or False negative.

if on the otherhand, we accept a hypothesis when it should be rejected we say that the error is type II or false positive.

Level of Significance

In testing a given hypothesis, the maximum probability with which we would be willing to risk a Type I error is called the level of significance of the test.

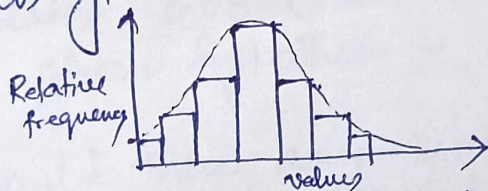
In practical real life problem we keep the

level of significance as 0.05 or 0.01.

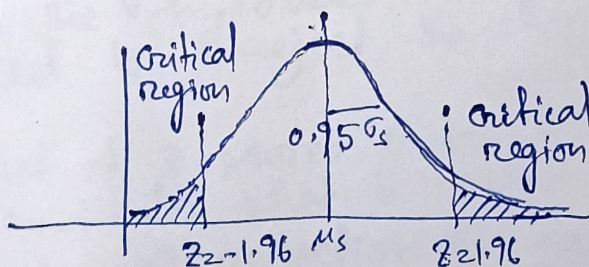
It means that if a 0.05 or 5% level of significance is chosen in designing a test of hypothesis, then there are about 5 chances in 100 that we would reject the hypothesis when it should be accepted, i.e. when ever the null hypothesis is true we are about 95% confident that we would make the right decision. In such cases we say that the hypothesis has been rejected at a 0.05 level of significance, which means that we could be wrong with probability 0.05.

Let us now take a model example to clarify the concepts.

Let's suppose that under a given hypothesis we have collected data and plotted the histogram and found as given below -



If we join the top tip of the bars, we get -



The mean of the distribution is μ_s and the standard deviation is σ_s .

$\therefore$ The distribution has the form -

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma_s} e^{-\frac{(x-\mu_s)^2}{2\sigma_s^2}}$$

Now we calculate the variable -

$$z_i = \frac{x_i - \mu_s}{\sigma_s}$$

This is called the z -score.

As indicated in the figure we find that if the z -value lies between -1.96 to $+1.96$ then the area under the normal curve is 0.95 .

On choosing a single sample if we find that its z -value is $-1.96 \leq z \leq 1.96$ then for it the hypothesis is true with probability 0.95 .

On the otherhand if we find that the z -score lies outside the range -1.96 to 1.96 we would conclude that such an event could happen with the probability 0.05 . (Total shaded area in the figure) if the given hypothesis ~~is~~ true.

We would then say that the z -score differed significantly from what would be expected under the hypothesis, and we would be inclined to reject the hypothesis.

The set of z -scores outside the range -1.96 to 1.96 constitutes what is called the critical region or region of rejection of the hypothesis or the region of significance.

The set of z -scores inside the range -1.96 to 1.96 could then be called the region of acceptance of the hypothesis or the region of non-significance.

On the basis of this model we can formulate the following decision rule.

- Reject the hypothesis at a 0.05 level of significance if the z -score lies outside the range -1.96 to 1.96 . This is called as the observed sample is significant at the 0.05 level.
 - Accept the hypothesis otherwise.
- For other levels of significance the limits of z -values changes.

Level of Significance:	0.10	0.05	0.01	0.005	0.002
Critical values of z	-1.645 $+1.645$	-1.96 $+1.96$	-2.58 $+2.58$	-2.81 $+2.81$	-3.08 $+3.08$

We will now consider numerical problems to clarify the concepts so far discussed.

Ex: 1 To test the hypothesis that a coin is fair, the following decision rule is adopted: (1) Accept the hypothesis if the number of heads in a single sample of 100 tosses is between 40 and 60 inclusive. (2) Reject the hypothesis otherwise.

- Find the probability of rejecting the hypothesis when it is actually correct.
- Interpret graphically the decision rule and the results of part (a).
- What conclusions would you draw if the sample of 100 tosses yielded 53 heads? 60 heads?
- Could you be wrong in your conclusion to (c)? Explain.

- a) To find the probability of getting 40 to 60 heads inclusive we consider that p be the probability of getting head in a single toss and $1-p$ is the probability of not getting head in a single toss, and $p = 1-p = \frac{1}{2}$

According to the binomial distribution the required probability is -

$${}^{100}C_{40} \left(\frac{1}{2}\right)^{40} \left(\frac{1}{2}\right)^{60} + {}^{100}C_{41} \left(\frac{1}{2}\right)^{41} \left(\frac{1}{2}\right)^{59} + \dots + {}^{100}C_{60} \left(\frac{1}{2}\right)^{60} \left(\frac{1}{2}\right)^{40}$$

${}^nC_n p^n (1-p)^{n-n}$

The mean of any binomial distribution is $\mu = np$

and the standard deviation $\sigma = \sqrt{np(1-p)}$

where $n \equiv$ total no of trial $= 100$ in this case.

$$\therefore \mu = 100 \times \frac{1}{2} = 50 \quad \sigma = \sqrt{100 \times \left(\frac{1}{2}\right) \left(\frac{1}{2}\right)} = \sqrt{100 \times \frac{1}{4}} = 5$$

$\therefore np$ and $nq = n(1-p)$ are both greater than 5, the binomial distribution can be approximated by normal distribution.

On a continuous scale, between 40 and 60 heads inclusive it is the same as between 39.5 and 60.5 heads.

in standard units.

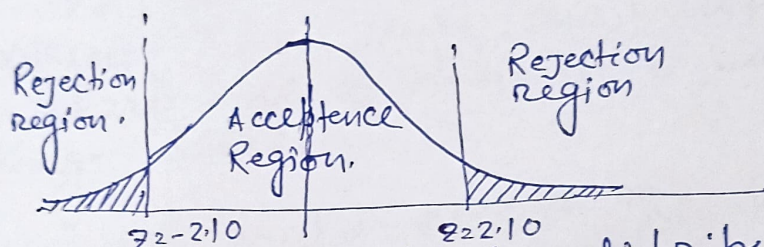
$$39.5 = \frac{39.5 - 50}{5} = -2.10 \quad \text{and} \quad 60.5 = \frac{60.5 - 50}{5} = 2.10$$

Required probability = area under normal curve
 $22-2.10$ and $22+2.10 = 2 \times 0.4821 = 0.9642$

$\therefore$ The probability of not getting between 40 and 60 heads inclusive if the coin is fair equals $1 - 0.9642 = 0.0358$.

$\therefore$ The probability of rejecting the hypothesis when it is correct is 0.0358.

b) The decision rule is illustrated in the figure below:



This shows the probability distribution of heads in 100 tosses of a fair coin.

If a single sample of 100 tosses yields a z -score between -2.10 and 2.10 we accept the hypothesis. Otherwise we reject the hypothesis and decide that the coin is not fair.

The error made in rejecting the hypothesis when it should be accepted is the Type-I error of the decision rule and the probability of making this ~~rule~~ error is 0.0358 and it is represented by the shaded area in the curve.

If a single sample of 100 tosses yields a number of heads whose z -score lies in the shaded regions, we could say that this z -score differed significantly from what

would be expected if the hypothesis were true. For this reason the total shaded area is called the level of significance of the decision rule. It equals 0.0358 in this case. We therefore speak of rejecting the hypothesis at a 0.0358 or 3.58% level of significance.

c) According to the decision rule, we would have to accept the hypothesis that the coin is fair in both cases. One might argue that if only one more head had been obtained, we could have rejected the hypothesis. This is what one must face when any sharp line of division is used in making decisions.

d) Yes. We could accept the hypothesis when it actually should be rejected, as would be the case, for example, when the probability of head is really 0.7 instead of 0.5.

The error made in rejecting accepting the hypothesis when it should be rejected is type II error.